

Sai Prasad Reddy Kukudala

kukudalasaiprasad@gmail.com | +1 405 413 6278 | LinkedIn | GitHub | Norman, OK | OPT-eligible (July 2026)

Professional Summary

Master's student in Computer Science at the University of Oklahoma (GPA: 3.82) graduating May 2026, with hands-on experience in ML/data projects, full-stack applications, and campus IT support. Focused on ML engineering, data science, GenAI/RAG pipelines, MLOps, and AI-focused backend/full-stack development with project work in React, FastAPI, PostgreSQL, Docker, LangChain, FAISS, XGBoost, LightGBM, SHAP, and Power BI. Seeking entry-level roles in ML Engineering, Data Engineering, Data Science, or AI-focused backend development.

Education

University of Oklahoma

M.S. in Computer Science — GPA: 3.82/4.00

Norman, OK

Aug 2024 – May 2026

Vardhaman College of Engineering

B.Tech in Computer Science — GPA: 8.72/10

Hyderabad, India

Nov 2020 – May 2024

Technical Skills

Languages: Python, SQL, Java, C, C++, JavaScript, HTML/CSS

Data / ML: Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, SHAP, TF-IDF, PyTorch, OpenCV

GenAI / NLP / Search: LangChain, Hugging Face, RAG, Embeddings, FAISS, Vector Search, pgvector

Backend / Full Stack: FastAPI, REST APIs, JWT Authentication, SQLAlchemy, PostgreSQL, Docker, Docker Compose, Nginx, MLflow, Uvicorn, Git

Frontend / BI / Tools: React.js, Vite, Axios, React Router, Recharts, Streamlit, Power BI, Linux, AWS, DuckDB, Hardhat, Android Studio

Projects

Smart IT Helpdesk & Asset Management System [\[GitHub\]](#)

Apr 2026 – Jun 2026

React.js · Vite · FastAPI · PostgreSQL · SQLAlchemy · Docker · Scikit-learn

- Built a full-stack IT support platform with JWT authentication, role-based access control, ticket workflows, asset inventory, user assignment, and analytics dashboards.
- Integrated TF-IDF and Logistic Regression to automatically classify support tickets into Hardware, Software, Network, Account Access, and Other from ticket title and description.
- Dockerized React, FastAPI, PostgreSQL, and Nginx services with Docker Compose and seeded demo data for reproducible local setup and portfolio demonstration.

Customer Churn Prediction with Explainable AI [\[GitHub\]](#)

Jun 2026

Python · SQL · Scikit-learn · XGBoost · SHAP · Streamlit · Power BI · Docker

- Built an end-to-end churn prediction system using telecom customer data, combining SQL-based EDA, classification modeling, explainability, and business dashboarding.
- Compared Logistic Regression, Random Forest, and XGBoost using accuracy, precision, recall, F1-score, and ROC-AUC; selected XGBoost for customer-level churn prediction.
- Integrated SHAP explanations, a Streamlit risk dashboard, Power BI executive reports, and Docker support to explain churn probability and top customer-specific churn reasons.

AI Job Match and Resume Analyzer using RAG [\[GitHub\]](#)

May 2026

Python · LangChain · Hugging Face · FAISS · PostgreSQL · Streamlit · Docker

- Built a RAG-powered resume analysis application that compares uploaded resumes against job descriptions, generating match scores, skill gaps, keyword suggestions, and project recommendations with downloadable reports.
- Implemented TF-IDF similarity scoring, Hugging Face embedding generation, and FAISS-based semantic retrieval with persistent PostgreSQL-backed analysis history across sessions.

Retail Demand Forecasting with MLOps [\[GitHub\]](#)

Apr 2026 – May 2026

Python · LightGBM · MLflow · FastAPI · PostgreSQL · Docker

- Built an end-to-end sales forecasting pipeline with time-series feature engineering, LightGBM model training, and MLflow experiment tracking across 20+ runs.
- Served real-time predictions through a FastAPI endpoint and stored all API outputs in PostgreSQL using a Dockerized multi-service architecture.

PostgreSQL-Based Content-Based Image Retrieval [\[GitHub\]](#)

Jan 2026 – May 2026

PostgreSQL · pgvector · HNSW · ResNet-50 · Python

- Built a PostgreSQL-native image similarity search system using pgvector and HNSW indexing with ResNet-50 embeddings on CIFAR-10 (50,000 images).
- Implemented adaptive filtered ANN retrieval and compact vector reranking, reducing filtered query latency from 891 ms to 203 ms — a 77% improvement.

Decentralized Freelance Marketplace [\[GitHub\]](#)

Mar 2025 – Apr 2025

Solidity · React.js · Hardhat · IPFS

- Built a blockchain-based freelance platform with Ethereum escrow payments, DAO-based dispute resolution, decentralized file storage via IPFS, and a tokenized reputation system.

Extended Iris Classification [\[GitHub\]](#)

Mar 2025 – Apr 2025

Python · k-NN · Neural Networks

- Built supervised learning models using k-NN and neural networks on an extended 21-feature Iris dataset to perform multi-class classification and compare model performance.

Maze Solver using Q-Learning and Monte Carlo [\[GitHub\]](#)

Jan 2025 – Mar 2025

Python · Reinforcement Learning

- Implemented Q-Learning and Monte Carlo agents to learn optimal maze paths via reward-based exploration, comparing convergence speed and policy quality across agent types.

WSafe - Women Safety Application

Sep 2023 – Jan 2024

Java · XML · Android Studio

- Developed an Android safety application with emergency support features to help users quickly share alerts and access safety assistance.
- Publication URL: doi.org/10.1007/978-3-031-81171-5_10

Automatic Essay Scoring — Deep Learning

Apr 2023 – Jun 2023

Python · LSTM · NLP

- Developed an LSTM-based model to score essays on a 1–10 scale by analyzing grammar, fluency, and semantic quality using sequence modeling on a public benchmark dataset.

Certifications

Google Cloud: Applying Machine Learning to your Data with Google Cloud; From Data to Insights with Google Cloud

Microsoft: Data Fundamentals DP-900; Power Platform Fundamentals PL-900